

From: "Neelesh malviya" <neeleash.malviya@arnasoftech.com>
Subject: eml with single PDF
To: "Neelesh malviya" <neeleash.malviya@arnasoftech.com>
Sent: January 27, 2026 1:11 PM (UTC+00:00)
Attached: indore.pdf

testtestName12 - Neelesh

description12 : Detaled discription of what is beeing requested

subject: Subjeckt of the requesst

Email122: test@mailinator.com

Starewte :MP

I am currntly lives in : Indore

Addres :indore bajrang nagar

SQL Server Interview Questions and Answers (4+ Years Experience)

Core SQL Concepts

1. What is the difference between WHERE and HAVING?

- **WHERE** filters rows before grouping, **HAVING** filters groups after **GROUP BY**.

```
SELECT Department, COUNT(*)  
FROM Employees  
WHERE IsActive = 1  
GROUP BY Department  
HAVING COUNT(*) > 5;
```

2. What is a CTE, and how is it different from a subquery?

- CTEs improve readability and allow recursion. Subqueries are inline.

```
WITH SalesCTE AS (  
    SELECT SalesPersonID, SUM(SalesAmount) AS TotalSales  
    FROM Sales  
    GROUP BY SalesPersonID  
)  
SELECT * FROM SalesCTE WHERE TotalSales > 10000;
```

3. How do you delete duplicate rows but keep one?

```
WITH CTE AS (  
    SELECT *, ROW_NUMBER() OVER (PARTITION BY Name, DOB ORDER BY ID) AS rn  
    FROM Employees  
)  
DELETE FROM CTE WHERE rn > 1;
```

4. JOIN types differences:

- **INNER JOIN** : Common records.
- **LEFT JOIN** : All from left + matching right.
- **RIGHT JOIN** : All from right + matching left.
- **FULL OUTER JOIN** : All records with NULLs where no match.

5. UNION vs UNION ALL

- **UNION** : Removes duplicates.

- `UNION ALL` : Keeps duplicates (faster).
-

Performance and Optimization

6. How to identify slow queries?

- Use SQL Profiler, Extended Events, DMVs (`sys.dm_exec_query_stats`).

7. What is an execution plan?

- A visual/text plan showing how SQL Server executes a query. Use `SET SHOWPLAN_XML ON` .

8. What are indexes?

- Indexes speed up searches.
- Clustered: Sorts the table.
- Non-clustered: Separate structure with pointer to table.

9. What is a covering index?

- An index that includes all columns required by the query (no need to access the table).

10. Optimization for ``?

- Use Full-Text Search instead of `LIKE` with wildcards at the beginning.
-

Stored Procedures and Functions

11. Scalar vs Table-Valued Functions vs Stored Procedures

- Scalar: Returns one value.
- TVF: Returns a table.
- SP: Executes logic, can return multiple result sets.

12. Can SP return a value?

- Yes, via `RETURN` or output parameters.

13. Pros/Cons of SPs vs inline SQL

- SP Pros: Better security, reusability, caching.
- Cons: Can get complex and hard to version.

14. Error handling in SPs:

```
BEGIN TRY
  -- SQL logic
END TRY
BEGIN CATCH
  -- Error logic
END CATCH
```

Transactions and Concurrency

15. What is a transaction? What are ACID properties?

- Atomicity, Consistency, Isolation, Durability.

16. Isolation Levels:

- Read Uncommitted, Read Committed (default), Repeatable Read, Snapshot, Serializable.

17. Deadlock detection and resolution:

- SQL Server picks a victim.
- Use SQL Profiler, sys.dm_tran_locks, proper indexing, and TRY...CATCH.

18. ** vs **

- Both are same. Allow dirty reads. Can lead to inconsistent results.

Advanced SQL Server Topics

19. Indexed Views

- Persisted on disk. Fast reads. Restrictions apply.

20. Parallelism

- SQL splits query into threads using multiple CPUs. Controlled via MAXDOP.

21. Window Functions:

```
SELECT Name, Salary, RANK() OVER (ORDER BY Salary DESC) AS SalaryRank
FROM Employees;
```

22. PIVOT and UNPIVOT

```
-- PIVOT Example
SELECT * FROM (
    SELECT Year, Quarter, Revenue FROM Sales
) AS SourceTable
PIVOT (
    SUM(Revenue) FOR Quarter IN ([Q1], [Q2], [Q3], [Q4])
) AS PivotTable;
```

23. Table Variables vs Temp Tables vs CTEs

- Table variables: Small datasets, memory-optimized.
- Temp tables: Larger, support indexes.
- CTE: Readable, recursive, not persisted.

Security and Access Control

24. Login vs User

- Login: Server-level.
- User: Database-level.

25. Grant/Revoke Example:

```
GRANT SELECT ON Employees TO User1;
REVOKE SELECT ON Employees FROM User1;
```

26. SQL Injection Prevention

- Use parameterized queries or stored procedures.

Real-World/Scenario-Based

27. Schema for Task Tracker

- Tables: Users, Tasks, Status, Comments.
- Use proper keys, indexes, and constraints.

28. Archiving Old Data

- Move to archive table.
- Use partitioning, batching (TOP (1000)), SQL Agent Jobs.

29. Migration Without Downtime

- Use replication, log shipping, or change data capture (CDC).

30. Schema Versioning

- Use tools like SSDT, Liquibase, Flyway.
- Keep scripts in version control.

Let me know if you want exercises, queries, or mock tests based on this list.